

KAI CHEUNG

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EDUCATION

University of California, Irvine | September 2023 - Present

B.S. Electrical Engineering, specialization in Digital Signal Processing
3rd Year with Junior Standing

Current GPA: 3.9

Expected 2027

Honors and Awards:

AICE Diploma, Received 2023

Cambridge Merit, Received 2021, 2022, 2023

PROJECT EXPERIENCE

MP3 Player

Hobby Project

Kirkland, WA

July 2025-Present

- **Designed and optimized a power management system** for a custom MP3 player, integrating a rechargeable LiPo battery with onboard charging and engineered solutions to increase battery life.
- **Engineered DAC power delivery** using a boost converter and MOSFET switching, addressing challenges with stable 5V regulation from a 3.7V supply.

Personal Music Production Project | DockKai

Passion Project

2019 - Present

- **Created unique experimental sounds** rooted in electronic music, drawing from a dearth of genres.
- **Explored the intersection of music and engineering** by studying synthesizers, recording hardware, and sound design techniques.
- **Applied fundamental synthesis methods** to design and produce original soundscapes and effects.

[Personal Keyboard](#)

Hobby Project

Kirkland, WA

June 2025-July 2025

- **Designed and fabricated** a fully wireless split ergonomic mechanical keyboard (“splitty”) by creating custom PCBs in KiCad, modeling the case and plate in Fusion 360, and sourcing/managing all components through online vendors.
- **Implemented custom firmware** using ZMK with device tree configurations, key matrix definitions, and multi-layer keymaps, enabling Bluetooth communication between two microcontrollers and seamless pairing with host devices.

Robot Arm | Zotbotics@UCI FABWorks

Club Project

Irvine, CA

January 2025 - June 2025

- **Assembled and wired a robotic arm**, configuring and tuning stepper motors with appropriate microstepping for smooth, precise motion control.
- **Developed Arduino firmware** to execute a rock-paper-scissors sequence on button press, unifying stepper and servo control into a single program.
- **Engineered a stable power system** using a 24 V supply with 12 V and 5 V buck converters to safely power microcontrollers and servos, ensuring reliable operation.

iPod Refurbishment Project | IEEE@UCI Lab

Irvine, CA

Hobby Project

Completed January 2025

- **Upgraded iPod Video** (5.5th Gen) with 2000mAh battery solution and flash storage.
- **Added Bluetooth Functionality** using generic 3rd party Bluetooth module and cleverly rewired existing iPod buttons for a seamless listening experience.
- Demonstrated proficiency in soldering and system optimization.

Guitar Pedals

Irvine, CA

Hobby Project

September 2024-June 2024

- **Built DIY fuzz distortion and reverb pedals** by ordering specialized PCBs, researching suitable electronic components and assembling by hand.
- **Reverse-engineered** a simple overdrive circuit, refining electronic filter design skills.

LEADERSHIP EXPERIENCE

WSSDA Conference Speaker

Hyatt Regency Ballroom, Bellevue, WA

Speaker

2021

- **Spoke at the annual Washington State School Directors Association conference**, participating in discourse between teachers and students.
- **Wrote a speech** analyzing personal experience with research skills-based education and responded to live questions and criticism.

PROFESSIONAL AFFILIATIONS

Institute of Electrical and Electronics Engineers @UC Irvine (IEEE@UCI)

2023-Present

ZiMS @UC Irvine (Zotbotics member)

2024-Present

PROFESSIONAL EXPERIENCE

Pedalheads

6046 West Lake Sammamish Pkwy NE, Redmond, WA

Bike Instructor

June 2024 – August 2024

- **Coached children through progressive riding lessons**, focusing on self-awareness, skill development, and safety abilities.
- **Developed leadership, collaboration, and problem-solving skills** by managing a 2:10 instructor-to-student ratio.
- **Prepared for and addressed various challenges under pressure**, collaborating with coworkers to ensure student safety.

Cyan Swim

6601 132nd Ave NE, Kirkland, WA

Swim Instructor

January 2023 – August 2023

- **Coached children individually**, teaching foundational swimming skills up to competitive swim team strokes. Proudly taught several students from zero experience to swimming alone, without help.
- **Led and planned swim team practices**, fostering an environment for improvement of all swimming skills for the team.

SKILLS

Technical Skills: KiCad, Fusion, Microsoft Office, Datasheets, Verilog, Music Production, Passive & Active Filter Design, SPICE Simulation, Arduino, Soldering

Soft Skills: Problem-solving Mindset, Adaptability, Interpersonal Skills, Collaboration, Public Speaking, Leadership